No.

Date:

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JA TTE

1. a. $4 + j7$

Tahanan (R) = 4 Ω dan reaktansi $(X_L) = 7\Omega$

$X_L = 2\pi f L$, dan Induktansi (L) =

$L = \frac{X_L}{2\pi f} = \frac{7}{2\pi(50)} = 0,0223\ H \Rightarrow 22,3\ mH$

b. $5 - j3$

Tahanan (R) = 5 Ω dan kapasitansi $(X_C) = 3$

$X_C = \frac{1}{2\pi f C}$, dan kapasitansi (C) =

$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(50)(3)} F = \frac{10^6}{2\pi(50)(3)} MF = 1061\ \mu F$

c. $6 \angle 45°$

$\Rightarrow 6\left(\frac{1}{2}\sqrt{2} + j\frac{1}{2}\sqrt{2}\right) = 3\sqrt{2} + j\frac{1}{2}\sqrt{2}$

Tahanan $R = 3\sqrt{2}$, reaktansi $(X_L) = 6\ \Omega$

$L = \frac{X_L}{2\pi f} = \frac{6}{3\pi(50)} = 0,0127\ H \Rightarrow 12,7\ mH$

d.

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2. $Z_1 = 2 + j6$ $\quad Z_2 = 5 - j2$

$\Rightarrow R = 2\,\Omega, X_L = 6\,\text{ohm}$ $\quad \Rightarrow R = 5\,\text{ohm}, X_C = 2\,\text{ohm}$

$L = \frac{6}{2\pi f} = \frac{6}{2\pi(50}$ $= 0{,}0191\,H$ $\quad C = \frac{1}{2\pi(50(20)}$ $= 0{,}00159\,F$

$\Rightarrow$ Jika dirangkai seri $= Z_{eq} = Z_1 + Z_2$

$= 2 + j6 + (5 - j2)$

$Z_{eq} = 7 + j4 \rightarrow 8{,}06 \angle$

$\sqrt{7^2 + 4^2} = \sqrt{65}$

$= 8{,}06$

$\arctan \frac{4}{7} = 29{,}72°$

- Arus $= \frac{V}{Z_{eq}} = \frac{100 \angle 0°}{8{,}06 \angle 29{,}72°} = 12{,}406 \angle -29{,}72°$

- Sudut phase : $\angle -29{,}72°$ artinya loading terhadap tegangan sebesar $-29{,}72°$

$\Rightarrow$ Jika dirangkai paralel : $Z_{eq} = \frac{Z_1 \times Z_2}{Z_1 + Z_2} = \frac{(2+j6)(5-j2)}{(2+j6)+(5-j2)}$

$= \frac{10 - j4 + j30 - 12j^2}{7 + j4}$

$= \frac{10 + 12 + j30 - j4}{7 + j4} = \frac{22 + j26}{7 + j4} = \frac{3{,}14 + j6}{7 + j4}$

$Z_{eq} = 7{,}218 \angle 64{,}215$

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Impedansi $z = 20 - j30 = 36,055 \angle$

3 a. Impendansi : $20 - j30$ ohm. ; Tegangan : 100 V, Frekuensi : 50 Hz

a. Dari z cartesian diperoleh tahanan $R = 20\ \Omega$

b. Dari z bahwa $X_C = 30$, kapasitansi :

$$C = \frac{1}{2\pi f X_C} = \frac{10^6}{2\pi(50)(30)} = 106,171\ \mu F = 106,17\ \Omega$$

c. Cari mod z dulu $= \sqrt{(20^2) + (-30)^2} = 36,055$

d. $I = \frac{V}{Z} = \frac{100 \angle 0^\circ}{36,055 \angle -56,31^\circ} = 2,7735 \angle 56,31^\circ$

d. Arus lagging terhadap tegangan sebesar 56,31

4 a. $X_L = 2\pi f L$, dengan asumsi $F = 50$ Hz, maka :

$= 2\pi 50\ (0,15)$

$= 47,1\ \Omega$

b. Impenclansi $z = 32 + j47,1$

c. Besar Arus

$$I = \frac{V}{Z} = \frac{240 \angle 0^\circ}{56,942 \angle 55,81^\circ} = 4,214 \angle -55,81^\circ$$

Arus leading terhadap tegangan sebesar $-55,81^\circ$

mod z

$* \sqrt{(32^2) + (47,1)^2}$

$= 56,942$

Cari Polar

$\tan^{-1} \frac{47,1}{32} = 55,81^\circ$

5. Tegangan = 200V ; F = 50 Hz ; R = 12 Ω ; L = 0,10 H

$X_L = 2\pi f L$

$= 2\pi(50)(0,1) = 31,4\ \Omega$

Impedansi $z = 12 + j31,4$

$$I = \frac{V}{Z} = \frac{200 \angle 0^\circ}{33,61 \angle 69,08^\circ} = 5,95 \angle -69,08^\circ$$

Arus leading terhadap tegangan sebesar $-69,08^\circ$

mod z

$* \sqrt{12^2 + 31,4^2} = 33,614$

* Cari dulu polar dari

$\tan^{-1} \frac{31,9}{12} = 69,08^\circ$

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